

Notes: De Loecker, Goldberg, Khandelwal & Pavcnik (Econometrica, 2016) Prices, Markups, and Trade Reform

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Prowess production and price data	4
3.2	Sector, industry, and product classification	4
3.3	Tariff data and input tariffs	4
3.4	India's trade liberalization	5
4	Models and empirical specifications	5
4.1	Product-level production technology	5
4.2	Markup formula	5
4.3	Marginal costs	5
4.4	Production-function estimation	6
4.5	Recovering input allocations in multi-product firms	6
4.6	Pass-through regression	6
4.7	Trade-reform regressions	6
4.8	Pro-competitive markup regression	7
5	Identification strategy	7
6	Tables and figures	8
6.1	Table I: Summary Statistics and Table II: Example of Sector, Industry, and Product Classifications	8
6.2	Table III: Average Output Elasticities and Table IV: Median Output Elasticities	8
6.3	Table V: Output Elasticities, Input Price Variation, and Sample Selection and Table VI: Markups by Sector	9
6.4	Figure 1: Markups, Marginal Costs, and Quantities and Figure 2: Markups, Costs, and Product Sales Share	10
6.5	Table VII: Pass-Through of Costs to Prices and Table VIII: Prices and Output Tariffs, Annual Regressions	11
6.6	Figure 3: Distribution of Prices and Figure 4: Distribution of Marginal Costs and Markups .	11
6.7	Table IX: Prices, Costs, and Markups and Tariffs and Table X: Pro-Competitive Effects of Output Tariffs	12
7	Interpretive synthesis	13
8	Short summary	13

9	Conclusion	14
9.1	Core conclusion	14
9.2	Economic intuition	14
9.3	Incremental contribution in one sentence	14
9.4	Takeaway	14

1 Big picture

- **Question.** How did India's trade liberalization affect factory-gate prices, markups, and marginal costs, and how much of cost reductions were passed through to consumers?
- **Core empirical challenge.** Prices, markups, and marginal costs are not usually observed together. Prices are observed in this setting, but markups and marginal costs must be recovered from production data.
- **Key setting.** The paper studies India's broad trade reform, especially the tariff declines after the 1991 balance-of-payments crisis. The reform cut both output tariffs and input tariffs.
- **Core methodological innovation.** The paper estimates product-level markups and marginal costs for multi-product firms without specifying demand, market structure, or a particular competitive model.
- **Main empirical result.** Trade liberalization reduced factory-gate prices, but the price declines were much smaller than the decline in marginal costs because firms raised markups.
- **Mechanism.** Input tariff reductions lowered marginal costs by making imported inputs cheaper and more varied. Firms did not fully pass these cost savings through to prices; instead, markups rose.
- **Pro-competitive effect.** Output tariff reductions put downward pressure on markups once the cost effect is controlled for. This pro-competitive effect is strongest for products with initially high markups.
- **Distributional implication.** Trade reform generated efficiency gains through lower marginal costs, but in the short run producers captured part of these gains through higher markups. Consumers received only incomplete pass-through.

2 Incremental contribution of the paper

- **Contribution 1: unified markup-cost framework for multi-product firms.** Earlier markup studies often estimated firm-level markups or imposed demand and conduct assumptions. This paper develops a production-based method for product-level markups and marginal costs for multi-product firms across many manufacturing industries.
- **Contribution 2: production-function methodology with quantity and price data.** The paper moves from revenue-based production-function estimation to quantity-based estimation. It explicitly addresses three biases: output price bias, input price bias, and the unobserved allocation of inputs across products inside multi-product firms.
- **Contribution 3: decomposition of trade reform effects.** Prior trade-and-markup studies focused mainly on output tariff reductions as a competition shock. This paper separately studies output tariff reductions and input tariff reductions, showing that input liberalization lowers costs while output liberalization disciplines markups.
- **Contribution 4: pass-through evidence in a developing-country trade reform.** The paper provides direct evidence on factory-gate price pass-through during a major liberalization episode. It shows that cost declines do not translate one-for-one into price declines because markups adjust endogenously.
- **Contribution 5: product-level evidence on multi-product firm behavior.** The paper uses within-firm product variation to show that products closer to the firm's core competence have lower marginal costs and higher markups, consistent with multi-product firm theory.
- **Contribution 6: policy interpretation.** The paper changes how trade liberalization should be evaluated. A reform can reduce distortions and costs while still delivering limited consumer price reductions if producers increase markups.

3 Data and measurement

3.1 Prowess production and price data

- The paper uses Prowess data from the Centre for Monitoring the Indian Economy. Prowess is a firm-level panel that tracks firms over time and reports detailed product-level information.
- The empirical setting covers Indian manufacturing over the trade liberalization period. The production-function and markup estimation use 1989–2003 data, while the trade-reform regressions focus on 1989–1997.
- The sample covers 11 two-digit manufacturing sectors, 1,970 firms in an average year, 1,047 single-product firms, and 1,400 products.
- Product-level information includes physical quantities and values, allowing the authors to compute unit prices for products.
- Because prices are observed at the product level, the paper can distinguish physical output from revenue. This is crucial for recovering true productivity, markups, and marginal costs.

Interpretation: The data are unusual because they contain price and quantity information for multiple products per firm. This is what makes product-level markup and marginal-cost estimation possible.

3.2 Sector, industry, and product classification

- Sectors are two-digit NIC categories, industries are four-digit NIC categories, and products are detailed product codes.
- Table II illustrates the hierarchy for basic metals: sector 27 contains industry 2710, which contains products such as pig iron, sponge iron, ferro alloys, and steel tubular products.
- The paper’s tariff variables are assigned at the industry level and then linked to products and firms through this classification system.

Interpretation: The hierarchy matters because the production-function estimation is organized by sector/product, while the trade-policy variation comes from output and input tariffs at industry level.

3.3 Tariff data and input tariffs

- Tariff data are available at the six-digit Harmonized System level from 1987 to 2001 and are concorded to India’s NIC industries.
- Output tariffs are industry-specific tariffs on final goods.
- Input tariffs are constructed by passing output tariffs through India’s 1993–1994 input-output matrix.

$$\tau_{it}^{input} = \sum_k a_{ki} \tau_{kt}^{output},$$

where a_{ki} is the share of industry k in the value of inputs used by industry i .

Interpretation: Output tariffs primarily capture changes in competitive pressure from imported final goods. Input tariffs capture changes in the cost of intermediate inputs.

3.4 India’s trade liberalization

- India undertook a major liberalization after the 1991 balance-of-payments crisis.
- Tariff reductions were large and heterogeneous across industries.
- The paper focuses on 1989–1997 because prior work shows tariff changes in this period were relatively uncorrelated with pre-reform industry characteristics.
- After 1997, tariff changes become more politically correlated with industry characteristics, so the baseline trade regressions stop before that period.

Interpretation: The paper’s identification is strongest in the early reform period, when tariff reductions are plausibly policy-driven rather than targeted to industry performance trends.

4 Models and empirical specifications

4.1 Product-level production technology

The paper begins with product-specific production:

$$Q_{fjt} = F_{jt}(V_{fjt}, K_{fjt})\Omega_{ft}.$$

Here Q_{fjt} is physical output, V_{fjt} is a vector of freely adjustable variable inputs, K_{fjt} is a vector of dynamic inputs, and Ω_{ft} is Hicks-neutral firm productivity.

- Technology is product-specific: a single-product and multi-product firm producing product j share the same physical technology for that product.
- Productivity is firm-specific and Hicks-neutral across all products within a firm.
- Inputs can be attributed to products, but the allocation shares are unobserved and must be recovered.
- Firms minimize short-run costs taking output and input prices as given.

Interpretation: The specification is deliberately production-based. It avoids specifying demand or conduct, but it requires assumptions on technology and input allocation.

4.2 Markup formula

Cost minimization for a variable input V^v yields the product-level markup:

$$\mu_{fjt} = \theta_{fjt}^v (\alpha_{fjt}^v)^{-1},$$

where θ_{fjt}^v is the output elasticity with respect to variable input v , and α_{fjt}^v is the variable input’s expenditure share in product sales.

Interpretation: The formula is the workhorse. To compute markups, the authors need two ingredients: output elasticities from the production function and product-level input expenditure shares. Both are nontrivial in multi-product firms.

4.3 Marginal costs

Once product-level prices and markups are known, marginal costs are recovered as:

$$mc_{fjt} = \frac{P_{fjt}}{\mu_{fjt}}.$$

Interpretation: This is why observing product prices is so powerful. Prices allow marginal costs to be recovered directly from markups.

4.4 Production-function estimation

- The paper estimates quantity-based production functions using single-product firms and firms observed before/after product switching.
- A translog functional form is used so output elasticities can vary across firms and over time.
- The estimation corrects for output price bias by using physical quantities, not deflated revenues.
- The estimation corrects for input price bias by allowing input prices to vary with output prices, market shares, tariffs, and other variables.
- The method uses GMM moments based on lagged prices, lagged market shares, tariffs, and lagged inputs to identify production and input-price parameters.

Interpretation: This methodology is one of the paper’s core contributions. Standard revenue production functions would confound price variation, quality, demand, markups, and productivity.

4.5 Recovering input allocations in multi-product firms

Multi-product firms report total input use but not the allocation of inputs across products. The paper recovers product-level input allocations using estimated technologies and the first-order conditions implied by cost minimization.

Interpretation: This step allows the authors to compute product-level input expenditure shares, and hence product-level markups, for multi-product firms. Without it, the product-level analysis would be impossible.

4.6 Pass-through regression

The pass-through regression relates log prices to log marginal costs:

$$\ln P_{fjt} = \lambda_{fj} + \beta \ln mc_{fjt} + \varepsilon_{fjt}.$$

Variances instrument marginal costs with input tariffs and lagged marginal costs.

Interpretation: If pass-through were complete, β would be close to one. The estimates are far below one, showing that cost shocks are only partly reflected in prices.

4.7 Trade-reform regressions

The main policy regression is:

$$y_{fjt} = \lambda_{fj} + \lambda_{st} + \lambda_1 \tau_{it}^{output} + \lambda_2 \tau_{it}^{input} + \eta_{fjt},$$

where y_{fjt} is log price, log marginal cost, or log markup. The regression includes firm-product fixed effects and sector-year fixed effects.

Interpretation: The regression compares the same firm-product pair over time while controlling for broad sector-year shocks. Output tariffs capture competition and input tariffs capture cost reductions.

4.8 Pro-competitive markup regression

To isolate the direct competition effect of output tariffs on markups, the paper controls flexibly for marginal costs:

$$\ln \mu_{fjt} = \lambda_{fj} + \lambda_{st} + \beta \tau_{it}^{output} + g(mc_{fjt}) + \eta_{fjt},$$

where $g(\cdot)$ is a second-order polynomial in marginal cost.

Interpretation: This matters because a markup regression without cost controls mixes pro-competitive effects with incomplete pass-through of cost reductions.

5 Identification strategy

- **Policy shock.** India's 1991 balance-of-payments crisis led to rapid trade liberalization. The exact tariff changes were not anticipated by firms before the reform.
- **Industry variation.** Tariff cuts varied across industries, and industries with initially higher tariffs experienced larger cuts.
- **Early-reform exogeneity.** Prior work shows tariff changes through 1997 were not correlated with pre-reform productivity, size, output growth, or capital intensity.
- **Firm-product fixed effects.** These absorb time-invariant differences in product quality, product type, and firm-product characteristics.
- **Sector-year fixed effects.** These absorb broad sectoral shocks and macroeconomic changes.
- **Input versus output channels.** Separating input and output tariffs is essential: input tariffs shift costs, while output tariffs shift competition.
- **Bootstrap standard errors.** The paper block bootstraps the full empirical procedure, including production-function estimation and markup/cost recovery.
- **Threats.** Remaining concerns include measurement error in product quantities/prices, input allocation assumptions, input market power, and tariff endogeneity outside the early reform window.

Interpretation: The strongest identifying variation is within firm-product changes over 1989–1997 induced by differential tariff declines across industries.

6 Tables and figures

6.1 Table I: Summary Statistics and Table II: Example of Sector, Industry, and Product Classifications

Read:

- Table I shows that the sample contains 1,970 firms and 1,400 products in an average year, across 11 manufacturing sectors.
- Chemicals and basic metals account for large shares of output, 26% and 16%, respectively.
- Table II illustrates the nested classification from sector to industry to product for basic metals.
- The classification example clarifies how product-level price and quantity data are mapped to industry-level tariffs.

Interpretation: These tables establish the data structure. The analysis needs enough products within industries to estimate product-level markups and enough firm-product observations to exploit within-product changes over time.

Economic message: The paper's empirical leverage comes from combining micro product data with macro policy shocks at the industry level.

PRICES, MARKUPS, AND TRADE REFORM

453

TABLE I
SUMMARY STATISTICS^a

Sector	Share of Sample	Single-Product		
	Output (1)	All Firms (2)	Firms (3)	Products (4)
15 Food products and beverages	9%	302	135	135
17 Textiles, apparel	10%	303	161	78
21 Paper and paper products	3%	77	56	32
24 Chemicals	26%	434	194	483
25 Rubber and plastic	5%	139	85	83
26 Nonmetallic mineral products	7%	110	74	60
27 Basic metals	16%	212	115	101
28 Fabricated metal products	2%	74	48	45
29 Machinery and equipment	7%	160	80	186
31 Electrical machinery and communications	5%	89	52	102
34 Motor vehicles, trailers	9%	71	47	95
Total	100%	1970	1047	1400

^aTable reports summary statistics for the average year in the sample. Column 1 reports the share of output by sector in the average year. Columns 2 and 3 report the number of firms and number of single-product firms manufacturing products in the average year. Column 4 reports the number of products by sector.

(a) Table I: Summary statistics.

TABLE II
EXAMPLE OF SECTOR, INDUSTRY, AND PRODUCT CLASSIFICATIONS^a

je	Description
	Basic metal industries (sector <i>s</i>)
	Manufacture of basic iron and steel (industry <i>i</i>)
	Products (<i>j</i>)
130101010000	Pig iron
130101020000	Sponge iron
130101030000	Ferro alloys
130106040800	Welded steel tubular poles
130106040900	Steel tubular structural poles
130106050000	Tube and pipe fittings
130106100000	Wires and ropes of iron and steel
130106100300	Stranded wire
	Casting of iron and steel (industry <i>i</i>)
	Products (<i>j</i>)
130106030000	Castings and forgings
130106030100	Castings
130106030101	Steel castings
130106030102	Cast iron castings
130106030103	Maleable iron castings
130106030104	S.G. iron castings

(b) Table II: sector, industry, and product classification.

6.2 Table III: Average Output Elasticities and Table IV: Median Output Elasticities

Read:

- Table III reports average output elasticities for labor, materials, and capital by sector.
- Returns to scale are close to one on average in many sectors, but average estimates can be distorted by outliers.
- Table IV reports median elasticities. Median materials elasticities are high and stable across sectors, consistent with materials being the flexible input used for markup recovery.
- Median returns to scale are generally close to one, making the production estimates economically plausible.

Interpretation: The output elasticities are the first ingredient in the markup formula. If these elasticities are implausible, the markups would be unreliable. The median table reassures the reader that the estimates are not driven by extreme observations.

Economic message: The production function estimates are not just intermediate steps; they determine the level and variation of recovered markups and marginal costs.

480 DE LOECKER, GOLDBERG, KHANDLWAL, AND PAVCNIK

TABLE III
AVERAGE OUTPUT ELASTICITIES, BY SECTOR^a

Sector	Observations in Production Function Estimation	Labor	Materials	Capital	Returns to Scale
	(1)	(2)	(3)	(4)	(5)
15 Food products and beverages	795	0.13 [0.17]	0.71 [0.22]	0.15 [0.14]	0.99 [0.28]
17 Textiles, apparel	1581	0.11 [0.02]	0.82 [0.04]	0.08 [0.08]	1.01 [0.06]
21 Paper and paper products	470	0.19 [0.12]	0.78 [0.10]	0.03 [0.05]	1.00 [0.06]
24 Chemicals	1554	0.17 [0.08]	0.79 [0.07]	0.08 [0.06]	1.03 [0.08]
25 Rubber and plastic	705	0.15 [0.39]	0.69 [0.29]	-0.02 [0.35]	0.82 [0.89]
26 Nonmetallic mineral products	633	0.16 [0.26]	0.67 [0.12]	-0.04 [0.40]	0.79 [0.36]
27 Basic metals	949	0.14 [0.09]	0.77 [0.11]	0.01 [0.06]	0.91 [0.18]
28 Fabricated metal products	393	0.18 [0.04]	0.75 [0.08]	0.03 [0.17]	0.96 [0.17]
29 Machinery and equipment	702	0.20 [0.08]	0.76 [0.05]	0.18 [0.05]	1.13 [0.14]
31 Electrical machinery and communications	761	0.09 [0.11]	0.78 [0.11]	-0.06 [0.22]	0.81 [0.28]
34 Motor vehicles, trailers	386	0.25 [0.26]	0.63 [0.20]	0.11 [0.20]	1.00 [0.25]

^aTable reports the output elasticities from the production function. Column 1 reports the number of observations for each production function estimation. Columns 2–4 report the average estimated output elasticity with respect to each factor of production for the translog production function for all firms. Standard deviations (not standard errors) of the output elasticities are reported in brackets. Column 5 reports the average returns to scale, which is the sum of the preceding three columns.

(a) Table III: average output elasticities.

PRICES, MARKUPS, AND TRADE REFORM 481

TABLE IV
MEDIAN OUTPUT ELASTICITIES, BY SECTOR^a

Sector	Labor	Materials	Capital	Returns to Scale
	(1)	(2)	(3)	(4)
15 Food products and beverages	0.12	0.75	0.20	1.09
17 Textiles, apparel	0.11	0.82	0.09	1.02
21 Paper and paper products	0.18	0.79	0.03	0.98
24 Chemicals	0.16	0.79	0.06	1.02
25 Rubber and plastic	0.21	0.75	0.04	1.03
26 Nonmetallic mineral products	0.18	0.69	0.04	0.88
27 Basic metals	0.14	0.78	0.02	0.96
28 Fabricated metal products	0.17	0.75	0.02	0.94
29 Machinery and equipment	0.17	0.75	0.16	1.08
31 Electrical machinery and communications	0.10	0.80	0.01	0.91
34 Motor vehicles, trailers	0.23	0.64	0.10	0.97

^aTable reports the median output elasticities from the production function. Columns 1–3 report the median estimated output elasticity with respect to each factor of production for the translog production function for all firms. Column 4 reports the median returns to scale.

(b) Table IV: median output elasticities.

6.3 Table V: Output Elasticities, Input Price Variation, and Sample Selection and Table VI: Markups by Sector

Read:

- Table V shows that ignoring input price variation produces implausible output elasticities, while ignoring sample selection has smaller effects.
- The table highlights input price bias as a serious problem in quantity-based production estimation.
- Table VI reports mean and median markups by sector. The mean and median across all sectors are 2.70 and 1.34, respectively.
- Sectoral markup variation is large: electrical machinery has a mean markup of 5.66, while paper products have a mean markup of 1.22.

Interpretation: Table V validates the methodological correction. Table VI shows that markups vary substantially across sectors and products, which makes it important to study the entire distribution rather than only average firm-level markups.

Economic message: Product-market power is heterogeneous. Trade reform can therefore affect firms and products very differently depending on baseline markups and cost structure.

to get a better sense of the plausibility of our estimates, we :

TABLE V
OUTPUT ELASTICITIES, INPUT PRICE VARIATION, AND SAMPLE SELECTION^a

Sector	Estimates Without Correcting for Input Price Variation				Estimates Without Correcting for Sample Selection			
	Labor (1)	Materials (2)	Capital (3)	Returns to Scale (4)	Labor (5)	Materials (6)	Capital (7)	Returns to Scale (8)
15 Food products and beverages	0.03	0.75	0.82	1.78	0.22	0.63	0.14	1.03
17 Textiles, apparel	-0.07	0.70	-0.07	0.52	0.11	0.83	0.09	1.03
21 Paper and paper products	-0.13	0.23	-0.19	-0.23	0.17	0.77	0.03	0.98
24 Chemicals	0.38	0.69	-0.72	0.26	0.16	0.79	0.04	0.99
25 Rubber and plastic	-0.10	0.30	-0.15	0.21	0.17	0.75	-0.05	0.94
26 Nonmetallic mineral products	0.08	0.64	0.81	1.50	0.12	0.71	0.11	0.93
27 Basic metals	-0.18	1.11	-0.33	0.69	0.12	0.80	0.02	0.94
28 Fabricated metal products	-1.17	-0.28	1.60	0.28	0.15	0.74	0.04	0.95
29 Machinery and equipment	-0.72	1.18	-0.50	-0.10	0.16	0.76	0.15	1.06
31 Electrical machinery and communications	-1.59	0.57	-0.13	-0.47	0.10	0.84	0.02	0.95
34 Motor vehicles, trailers	-0.23	-0.39	1.23	0.44	0.20	0.70	0.04	0.94

^aColumns 1-4 report the median output elasticities from production function estimations that do not account for input price variation. Columns 5-8 reports the median output elasticities from production function estimations that do account for sample selection (transition from single-product to multi-product firms).

(a) Table V: input price variation and sample selection.

TABLE VI
MARKUPS, BY SECTOR^a

Sector	Markups	
	Mean	Median
15 Food products and beverages	1.78	1.15
17 Textiles, apparel	1.57	1.33
21 Paper and paper products	1.22	1.21
24 Chemicals	2.25	1.36
25 Rubber and plastic	4.52	1.37
26 Nonmetallic mineral products	4.57	2.27
27 Basic metals	2.54	1.20
28 Fabricated metal products	3.70	1.36
29 Machinery and equipment	2.48	1.34
31 Electrical machinery and communications	5.66	1.43
34 Motor vehicles, trailers	4.64	1.39
Average	2.70	1.34

^aTable displays the mean and median markup by sector for the sample 1989–2003. The table trims observations with markups that are above and below the 3rd and 97th percentiles within each sector.

(b) Table VI: markups by sector.

6.4 Figure 1: Markups, Marginal Costs, and Quantities and Figure 2: Markups, Costs, and Product Sales Share

Read:

- Figure 1 shows markups increasing with quantity and marginal costs decreasing with quantity after demeaning by product.
- Figure 2 shows that within multi-product firms, products with larger sales shares have higher markups and lower marginal costs.
- These patterns are consistent with multi-product firm theory in which firms have a core competence product with low marginal cost and high markup.

Interpretation: The figures provide plausibility checks for the recovered markups and marginal costs. The estimates line up with theory even though the procedure does not impose a demand system or market-structure model.

Economic message: Product-level markup heterogeneity is economically meaningful inside firms: core products look more profitable and lower cost than peripheral products.

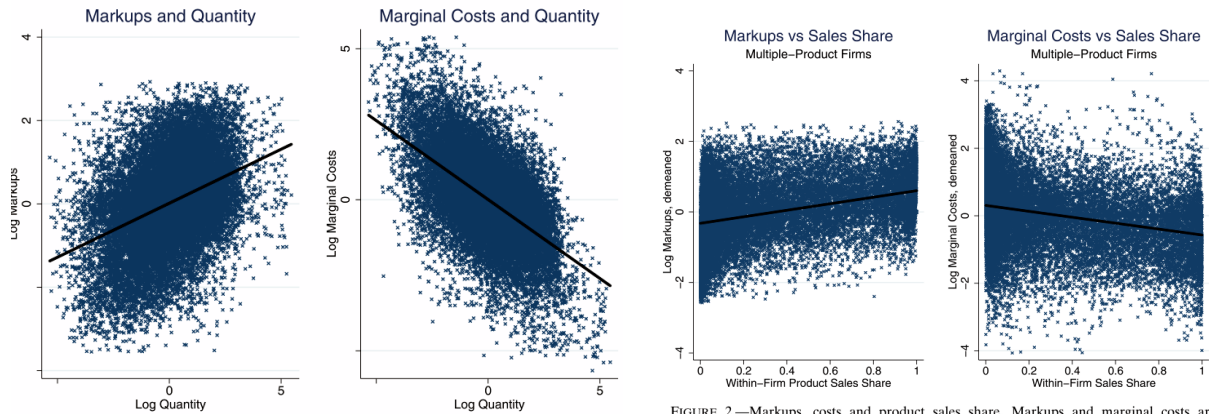


FIGURE 1.—Markups, marginal costs and quantities. Variables de-meaned by product. Markups, cost and quantity outliers are trimmed below and above 3rd and 97th percentiles.

(a) Figure 1: markups, marginal costs, and quantities.

FIGURE 2.—Markups, costs and product sales share. Markups and marginal costs are de-meaned by product-year and firm-year FEs. Markup and marginal cost outliers are trimmed below and above 3rd and 97th percentiles.

(b) Figure 2: markups, costs, and product sales share.

6.5 Table VII: Pass-Through of Costs to Prices and Table VIII: Prices and Output Tariffs, Annual Regressions

Read:

- Table VII estimates pass-through of marginal costs to prices. The OLS coefficient is 0.337, and IV estimates remain well below one.
- Low pass-through implies that shocks to marginal costs do not translate proportionally into factory-gate prices.
- Table VIII shows that output tariff reductions are associated with price declines: the output tariff coefficient is positive, so lower tariffs lower prices.
- The implied average price decline from output tariff liberalization alone is about 8.4–10.4%.

Interpretation: Table VII explains why price declines can be small even when costs fall a lot. Table VIII confirms the standard pro-competitive intuition that lower output tariffs lower prices.

Economic message: The price response combines two mechanisms: competition from output tariff reductions and incomplete pass-through of input-cost reductions.

DE LOECKER, GOLDBERG, KHANDELWAL, AND PAVCNİK

TABLE VII
PASS-THROUGH OF COSTS TO PRICES^a

	$\ln P_{fjt}$		
	(1)	(2)	(3)
$\ln mc_{fjt}$	0.337*** 0.041	0.305*** 0.084	0.406† 0.247
Observations	21,246	16,012	12,334
Within <i>R</i> -squared	0.27	0.19	0.09
Firm–product FEs	yes	yes	yes
Instruments	–	yes	yes
First-stage <i>F</i> -test	–	98	5

^aThe dependent variable is (log) price. Column 1 is an OLS regression on log marginal costs. Column 2 instruments marginal costs with input tariffs and lag marginal costs. Column 3 instruments marginal costs with input tariffs and two-period lag marginal costs. The regressions exclude outliers in the top and bottom 3rd percent of the markup distribution. All regressions include firm–product fixed effects. The regressions use data from 1989–1997. The standard errors are bootstrapped and are clustered at the firm level. Significance: †10.1 percent, *10 percent, **5 percent, ***1 percent.

(a) Table VII: cost pass-through to prices.

DE LOECKER, GOLDBERG, KHANDELWAL, AND PAVCNİK

TABLE VIII
PRICES AND OUTPUT TARIFFS, ANNUAL REGRESSIONS^a

	$\ln P_{fjt}$	
	(1)	(2)
$\tau_{it}^{\text{output}}$	0.136** 0.056	0.167*** 0.054
Within <i>R</i> -squared	0.00	0.02
Observations	21,246	21,246
Firm–product FEs	yes	yes
Year FEs	yes	no
Sector–year FEs	no	yes
Overall impact of trade liberalization	–8.4** 3.4	–10.4*** 3.3

^aThe dependent variable is a firm–product’s (log) price. Column 1 includes year fixed effects and Column 2 includes sector–year fixed effects. The regressions exclude outliers in the top and bottom 3rd percent of the markup distribution. All regressions include firm–product fixed effects and use data from 1989–1997. Standard errors are clustered at the industry level. The final row uses the average 62% decline in output tariffs from 1989–1997 to compute the mean and standard error of the impact of trade liberalization on prices. That is, for each column the mean impact is equal to the $-0.62 \times 100 \times$ (coefficient on output tariffs). Significance: *10 percent, **5 percent, ***1 percent.

(b) Table VIII: prices and output tariffs.

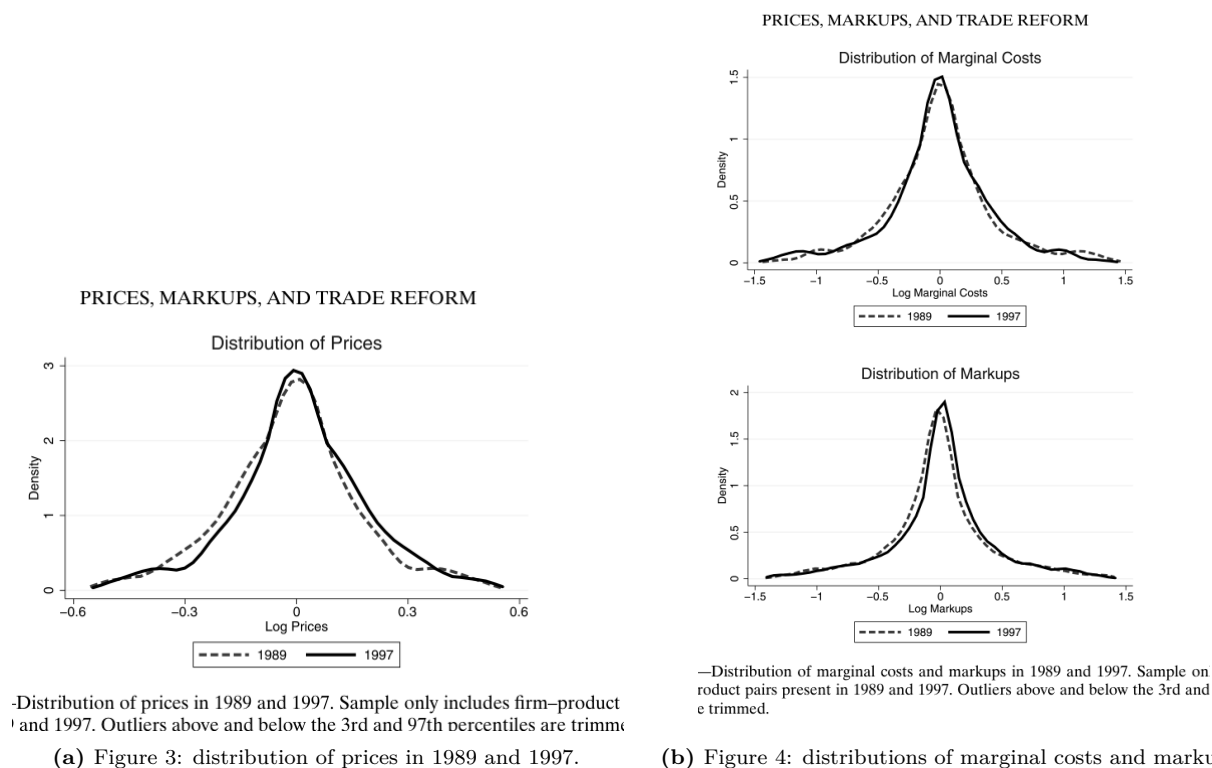
6.6 Figure 3: Distribution of Prices and Figure 4: Distribution of Marginal Costs and Markups

Read:

- Figure 3 shows the distribution of real prices in 1989 and 1997 for matched firm–product pairs; the distribution changes little.
- Figure 4 shows the marginal cost distribution shifts left while the markup distribution shifts right.
- The raw distributional evidence suggests that efficiency gains from lower costs were partly offset by higher markups.

Interpretation: These figures are the visual core of the paper. Prices look stable because two components move in opposite directions: marginal costs fall, while markups rise.

Economic message: Consumers saw only muted price declines even though firms experienced large cost reductions; producers captured part of the gains.



6.7 Table IX: Prices, Costs, and Markups and Tariffs and Table X: Pro-Competitive Effects of Output Tariffs

Read:

- Table IX decomposes the average trade-reform effect into prices, marginal costs, and markups.
- The average effect of the reform is a price decline of 18.1%, a marginal cost decline of 30.7%, and a markup increase of 12.6%.
- Input tariffs strongly affect marginal costs and markups: lower input tariffs reduce costs and are associated with higher markups.
- Table X controls flexibly for marginal costs. Conditional on costs, output tariff reductions lower markups, revealing the pro-competitive effect of output liberalization.
- The pro-competitive effect is stronger for products that initially had high markups.

Interpretation: Table IX explains the incomplete pass-through puzzle, while Table X isolates competition from cost pass-through. Reading only Table IX's markup column would be misleading because markup changes combine competition and cost shocks.

Economic message: Trade liberalization produced two opposing markup forces: output tariff cuts reduced market power, but input tariff cuts lowered costs and allowed firms to raise markups through incomplete pass-through.

TABLE IX
PRICES, COSTS, AND MARKUPS AND TARIFFS^a

	$\ln P_{fjt}$ (1)	$\ln mc_{fjt}$ (2)	$\ln \mu_{fjt}$ (3)
$\tau_{it}^{\text{output}}$	0.156***	0.047	0.109
	0.059	0.084	0.076
τ_{it}^{input}	0.352	1.160**	-0.807 [‡]
	0.302	0.557	0.510
Within <i>R</i> -squared	0.02	0.01	0.01
Observations	21,246	21,246	21,246
Firm-product FEs	yes	yes	yes
Sector-year FEs	yes	yes	yes
Overall impact of trade liberalization	-18.1**	-30.7**	12.6
	7.4	13.4	11.9

^aThe dependent variable is noted in the columns. The sum of the coefficients from the markup and marginal costs regression equals their respective coefficient in the price regression. The regressions exclude outliers in the top and bottom 3rd percent of the markup distribution, and include firm-product fixed effects and sector-year fixed effects. The final row uses the average 62% and 24% declines in output and input tariffs from 1989–1997, respectively, to compute the mean and standard error of the impact of trade liberalization on each performance measure. That is, for each column the mean impact is equal to the $-0.62 \times 100 \times$ (coefficient on output tariff) $\pm 0.24 \times 100 \times$ (coefficient on input tariff). The regressions use data from 1989–1997. The table reports the bootstrapped standard errors that are clustered at the industry level. Significance: [‡]11.3 percent, *10 percent, **5 percent, ***1 percent.

^bThe regressions exclude outliers in the top and bottom third percent of the markup distribution.

(a) Table IX: prices, costs, markups, and tariffs.

TABLE X
PRO-COMPETITIVE EFFECTS OF OUTPUT TARIFFS^a

	$\ln \mu_{fjt}$			
	(1)	(2)	(3)	(4)
$\tau_{it}^{\text{output}}$	0.143***	0.150**	0.129**	0.149**
	0.050	0.062	0.052	0.062
$\tau_{it}^{\text{output}} \times \text{Top}_{fp}$			0.314**	0.028
			0.134	0.150
Within <i>R</i> -squared	0.59	0.65	0.59	0.65
Observations	21,246	16,012	21,246	16,012
Second-order polynomial of marginal cost	yes	yes	yes	yes
Firm-product FEs	yes	yes	yes	yes
Sector-year FEs	yes	yes	yes	yes
Instruments	no	yes	no	yes
First-stage <i>F</i> -test	—	8.6	—	8.6

^aThe dependent variable is (log) markup. All regressions include firm-product fixed effects, sector-year fixed effects and a second-order polynomial of marginal costs (these coefficients are suppressed and available upon request). Columns 2 and 4 instrument the second-order polynomial of marginal costs with second-order polynomial of lag marginal costs and input tariffs. Columns 3 interacts output tariffs and the second-order marginal cost polynomial with an indicator if a firm-product observation was in the top 10 percent of its sector's markup distribution when it first appears in the sample. The regressions exclude outliers in the top and bottom 3rd percent of the markup distribution. The table reports the bootstrapped standard errors that are clustered at the industry level. Significance: *10 percent, **5 percent, ***1 percent.

(b) Table X: pro-competitive output tariff effects.

7 Interpretive synthesis

- **Prices alone are insufficient.** A small price decline can hide large cost declines and large markup increases.
- **Input tariffs matter.** Trade reforms are not only about final-good competition. Imported inputs can substantially lower marginal costs.
- **Markups are endogenous.** Firms change markups in response to both competition and cost shocks. Treating markups as constant misses a central distributional channel.
- **Consumer gains can be muted.** If firms increase markups when costs fall, consumers receive only partial pass-through of efficiency gains.
- **Method matters.** Revenue-based productivity measures cannot reveal this decomposition because they confound physical productivity, prices, markups, and demand.

8 Short summary

- The paper studies India's trade liberalization using Prowess product-level price and quantity data.
- It develops a production-based method for estimating product-level markups and marginal costs for multi-product firms.
- The method corrects output price bias, input price bias, and unobserved input allocation across products.
- Output tariffs and input tariffs have different economic roles: output tariffs affect competition; input tariffs affect marginal costs.
- Factory-gate prices decline after reform, but not as much as marginal costs.
- Input tariff reductions lower marginal costs by about 30.7% on average.
- Markups rise by about 12.6% on average, offsetting nearly half of the cost decline.

- Conditional on marginal costs, output tariff reductions reduce markups, especially for initially high-markup products.
- Producers captured part of the gains from input liberalization through higher markups, so consumer price gains were moderated.

9 Conclusion

9.1 Core conclusion

De Loecker, Goldberg, Khandelwal, and Pavcnik show that trade liberalization can generate large marginal-cost reductions without equally large price reductions. In India, input tariff liberalization lowered costs, but firms raised markups and passed through only part of the efficiency gains to consumers.

9.2 Economic intuition

Lower input tariffs make production cheaper. If markets were perfectly competitive or markups were constant, prices would fall roughly with costs. Instead, firms with market power adjust markups. Output tariff cuts discipline markups through competition, but input tariff cuts generate cost savings that firms can partly keep.

9.3 Incremental contribution in one sentence

The paper's distinctive contribution is to jointly estimate product-level prices, markups, and marginal costs in multi-product firms and to use this decomposition to show that trade reform created large cost savings but incomplete consumer pass-through.

9.4 Takeaway

Trade liberalization should be evaluated through prices, costs, and markups together. Studying any one of them in isolation can misstate who gains from reform and how much competition actually increases.